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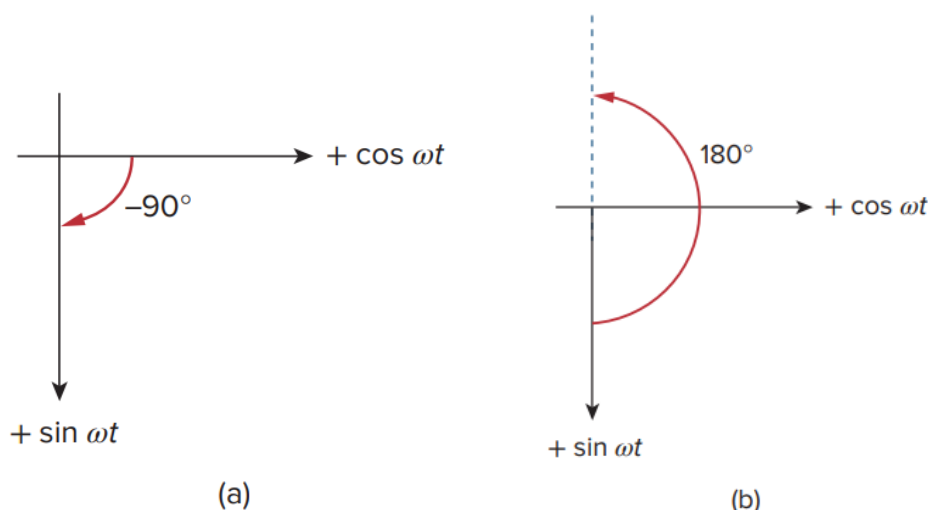
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Chapter 1

AC circuit analysis

1.1 Sinusoids and Phasors

1.1.1 Graphical representation of sinusoid signals



1. In AC circuits, **time-domain** sinusoidal voltage and currents are represented by a sine or a cosine function, so called a **sinusoid**.
2. The graphical conversion from sine form to cosine form or vice versa (Fig. 9.3):
 - $\cos \omega t$ defines the reference phase.
 - cosine to sine (1) $\cos(\omega t - 90^\circ) = + \sin \omega t$
 - cosine to sine (2) $\cos(\omega t + 90^\circ) = - \sin \omega t$
 - sine to cosine (1) $\sin(\omega t + 90^\circ) = + \cos \omega t$
 - sine to cosine (2) $\sin(\omega t - 90^\circ) = - \cos \omega t$
 - sign conversion (1) $\cos(\omega t \pm 180^\circ) = - \cos \omega t$
 - sign conversion (2) $\sin(\omega t \pm 180^\circ) = - \sin \omega t$

3. **Example:** Graphical approach to obtaining the phase angle between two sinusoids. Calculate the phase angle between $v_1 = -10 \cos(\omega t + 50^\circ)$ and $v_2 = 12 \sin(\omega t - 10^\circ)$. State which sinusoid is leading.

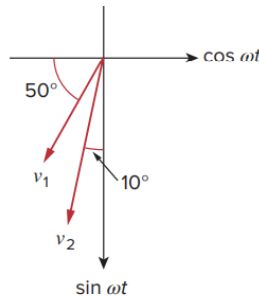


Figure 9.5
For Example 9.2.

Solution. It is clear from Fig. 9.5 that v_2 leads v_1 by 30° (or equivalently, v_1 lags v_2 by 30°).

4. **Example:** Adding two sinusoids of the same frequency.

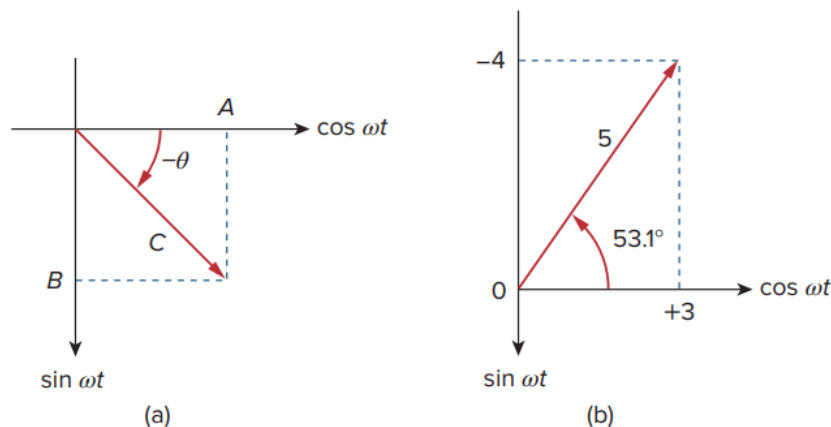


Figure 9.4

(a) Adding $A \cos \omega t$ and $B \sin \omega t$, (b) adding $3 \cos \omega t$ and $-4 \sin \omega t$.

To add $A \cos \omega t$ and $B \sin \omega t$, we note that A is the magnitude of $\cos \omega t$, while B is the magnitude of $\sin \omega t$, as shown in Fig. 9.4(a). The magnitude and argument of the resultant sinusoid in cosine form is readily obtained from the triangle. Thus,

$$A \cos \omega t + B \sin \omega t = C \cos(\omega t - \theta)$$

where

$$C = \sqrt{A^2 + B^2}, \quad \theta = \tan^{-1} \frac{B}{A}$$

For example, we may add $3 \cos \omega t$ and $-4 \sin \omega t$ as shown in Fig. 9.4(b) and obtain

$$3 \cos \omega t - 4 \sin \omega t = 5 \cos(\omega t + 53.1^\circ)$$

1.1.2 Phasors : Working in the frequency domain

1. A **phasor** is a complex number that represents the amplitude and phase of a sinusoid.
2. Given a sinusoid $v(t) = V_m \cos(\omega t + \phi)$,

$$\begin{aligned}
 v(t) &= V_m \cos(\omega t + \phi) \\
 &= \operatorname{Re} [V_m e^{j(\omega t + \phi)}] \\
 &= \operatorname{Re} [\mathbf{V} e^{j\omega t}] \\
 \text{where } \mathbf{V} &= V_m e^{j\phi} = V_m \angle \phi
 \end{aligned}$$

This $\mathbf{V}$ is the **phasor** representation of the time-domain sinusoids $v(t)$. The following table provides the sinusoid-phasor transformation for the most common voltage and current sinusoids.

TABLE 9.1

Sinusoid-phasor transformation.

Time domain representation	Phasor domain representation
$V_m \cos(\omega t + \phi)$	$V_m \angle \phi$
$V_m \sin(\omega t + \phi)$	$V_m \angle \phi - 90^\circ$
$I_m \cos(\omega t + \theta)$	$I_m \angle \theta$
$I_m \sin(\omega t + \theta)$	$I_m \angle \theta - 90^\circ$

3. The way of looking at the complex sinusoids $v(t) = \operatorname{Re} [\mathbf{V} e^{j\omega t}]$, called a *sinor*, is to view it as a rotating phasor at the frequency ω , as shown in Fig. 9.7:

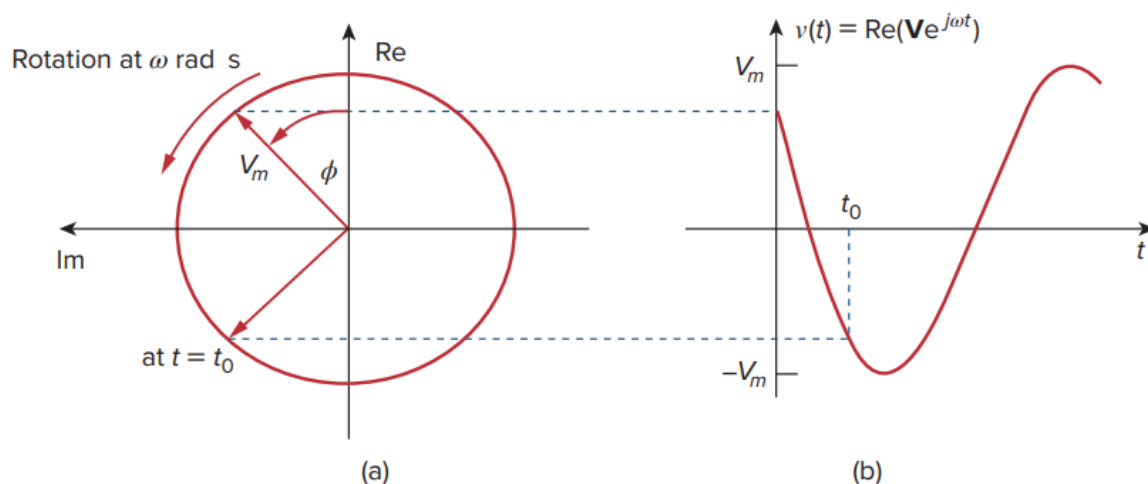


Figure 9.7

Representation of $\mathbf{V} e^{j\omega t}$: (a) sinor rotating counterclockwise, (b) its projection on the real axis, as a function of time.

4. Because a phasor has magnitude and phase (“direction”), it behaves as a **vector** and is printed in boldface. A graphical representation of phasors is known as a phasor diagram, whose example is provided below as in Fig. 9.8 for $\mathbf{V} = V_m \angle \phi$ and $\mathbf{I} = I_m \angle -\theta$

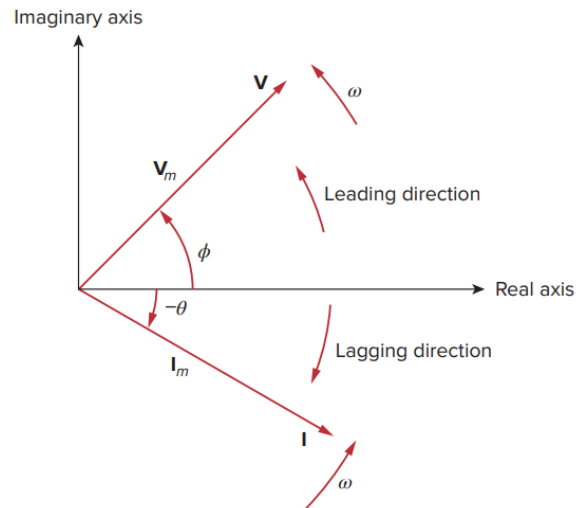


Figure 9.8

A phasor diagram showing $\mathbf{V} = V_m \angle \phi$ and $\mathbf{I} = I_m \angle -\theta$.

Note that the frequency (or time) factor $e^{j\omega t}$ is **suppressed**, and the frequency is not explicitly shown in the phasor domain representation because ω is constant. However, the response depends on ω . For this reason, the phasor domain is also known as the **frequency domain**.

5. Given a sinusoid $v(t) = V_m \cos(\omega t + \phi)$,

$$\frac{dv}{dt} = -\omega V_m \sin(\omega t + \phi) = \omega V_m \cos(\omega t + \phi + 90^\circ)$$

$$= \text{Re}(\omega V_m e^{j\omega t} e^{j\phi} e^{j90^\circ}) = \text{Re}(j\omega V e^{j\omega t})$$

$$\frac{dv}{dt} \quad (\text{Time domain}) \quad \Longleftrightarrow \quad j\omega \mathbf{V} \quad (\text{Phasor domain})$$

Similarly,

$$\int v \, dt \quad (\text{Time domain}) \quad \Longleftrightarrow \quad \frac{\mathbf{V}}{j\omega} \quad (\text{Phasor domain})$$

These relationship allows the replacement of a derivative with respect to time with multiplication of $j\omega$ in the phasor domain, whereas an integral with respect to time is replaced with division by $j\omega$ in the phasor domain. These properties of the phasor are useful in finding the steady-state solution, which does not require knowing the initial values of the variable involved.

6. **Example.** Using the phasor approach, determine the current $i(t)$ in a circuit described by the integrodifferential equation

$$4i + 8 \int i \, dt - 3 \frac{di}{dt} = 50 \cos(2t + 75^\circ)$$

Solution:

We transform each term in the equation from time domain to phasor domain. Keeping Eqs. (9.27) and (9.28) in mind, we obtain the phasor form of the given equation as

$$4\mathbf{I} + \frac{8\mathbf{I}}{j\omega} - 3j\omega\mathbf{I} = 50\angle 75^\circ$$

But $\omega = 2$, so

$$\mathbf{I}(4 - j4 - j6) = 50\angle 75^\circ$$

$$\mathbf{I} = \frac{50\angle 75^\circ}{4 - j10} = \frac{50\angle 75^\circ}{10.77\angle -68.2^\circ} = 4.642\angle 143.2^\circ \text{ A}$$

Converting this to the time domain,

$$i(t) = 4.642 \cos(2t + 143.2^\circ) \text{ A}$$

Keep in mind that this is only the steady-state solution, and it does not require knowing the initial values.

7. **Remarks.** The differences between $v(t)$ and $\mathbf{V}$ should be emphasized:
- (a) $v(t)$ is the *instantaneous or time domain* representation, while $\mathbf{V}$ is the *frequency or phasor domain* representation.
 - (b) $v(t)$ is time dependent, while $\mathbf{V}$ is not. (This fact is often forgotten by students.)
 - (c) $v(t)$ is always real with no complex term, while $\mathbf{V}$ is generally complex.
8. Finally, we should bear in mind that phasor analysis applies **only when frequency is constant**; it applies in manipulating two or more sinusoidal signals only if they are of **the same frequency**. Summing two sinusoids of the same frequency is illustrated in the following example.

Example: Given $i_1(t) = 4\cos(\omega t + 30^\circ)$ A and $i_2(t) = 5\sin(\omega t - 20^\circ)$ A, find their sum.

Solution:

Here is an important use of phasors—for summing sinusoids of the same frequency. Current $i_1(t)$ is in the standard form. Its phasor is

$$\mathbf{I}_1 = 4\angle 30^\circ$$

We need to express $i_2(t)$ in cosine form. The rule for converting sine to cosine is to subtract 90° . Hence,

$$i_2 = 5\cos(\omega t - 20^\circ - 90^\circ) = 5\cos(\omega t - 110^\circ)$$

and its phasor is

$$\mathbf{I}_2 = 5\angle -110^\circ$$

If we let $i = i_1 + i_2$, then

$$\begin{aligned}\mathbf{I} &= \mathbf{I}_1 + \mathbf{I}_2 = 4\angle 30^\circ + 5\angle -110^\circ \\ &= 3.464 + j2 - 1.71 - j4.698 = 1.754 - j2.698 \\ &= 3.218\angle -56.97^\circ \text{ A}\end{aligned}$$

1.1.3 Phasor relationships for circuit elements

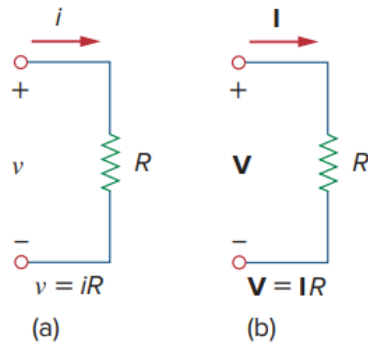


Figure 9.9

Voltage-current relations for a resistor in the: (a) time domain, (b) frequency domain.

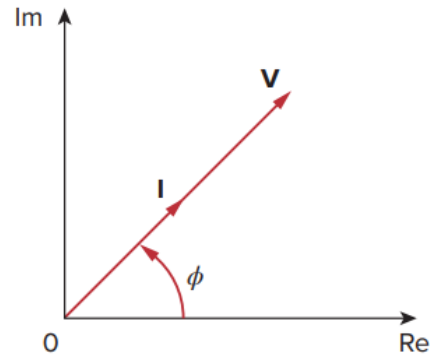


Figure 9.10

Phasor diagram for the resistor.

1. We begin with the resistor. If the current through a resistor R is $i = I_m \cos(\omega t + \phi)$, the voltage across it is given by Ohm's law as $v = iR = RI_m \cos(\omega t + \phi)$.

The phasor form of this voltage is $\mathbf{V} = RI_m \angle \phi$. But the phasor representation of the current is $\mathbf{I} = I_m \angle \phi$. Hence,

$$\mathbf{V} = R\mathbf{I}$$

showing that the voltage-current relation for the resistor in the phasor domain continues to be Ohm's law, as in the time domain.

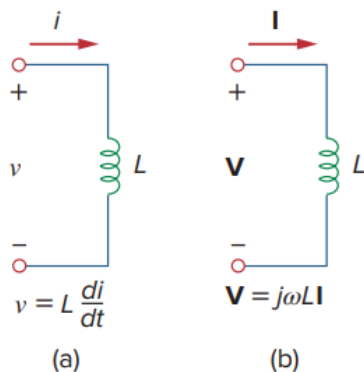


Figure 9.11

Voltage-current relations for an inductor in the: (a) time domain, (b) frequency domain.

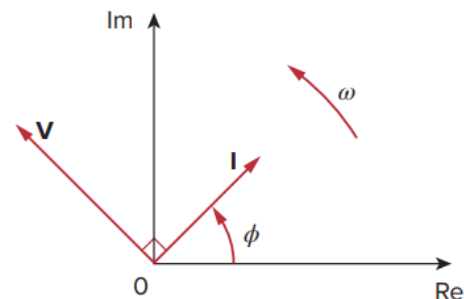


Figure 9.12

Phasor diagram for the inductor; $\mathbf{I}$ lags $\mathbf{V}$.

2. For the inductor L , assume the current through it is $i = I_m \cos(\omega t + \phi)$. The voltage across the inductor is

$$v = L \frac{di}{dt} = -\omega LI_m \sin(\omega t + \phi) = \omega LI_m \cos(\omega t + \phi + 90^\circ)$$

which transforms to the phasor

$$\mathbf{V} = \omega LI_m e^{j(\phi + 90^\circ)} = \omega LI_m e^{j\phi} e^{j90^\circ} = \omega LI_m \angle (\phi + 90^\circ) \quad (9.34)$$

But $I_m \angle \phi = \mathbf{I}$, and $e^{j90^\circ} = j$. Thus,

$$\mathbf{V} = j\omega L \mathbf{I} \quad (9.35)$$

showing that the voltage has a magnitude of $\omega L I_m$ and a phase of $\phi + 90^\circ$. The voltage and current are 90° out of phase. Specifically, the current lags the voltage by 90° .

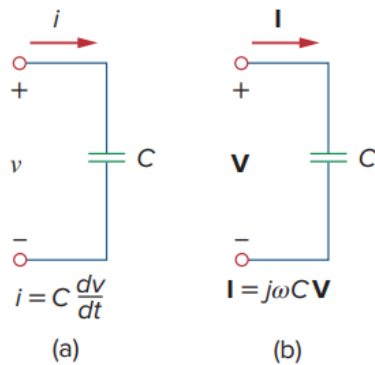


Figure 9.13

Voltage-current relations for a capacitor in the: (a) time domain, (b) frequency domain.

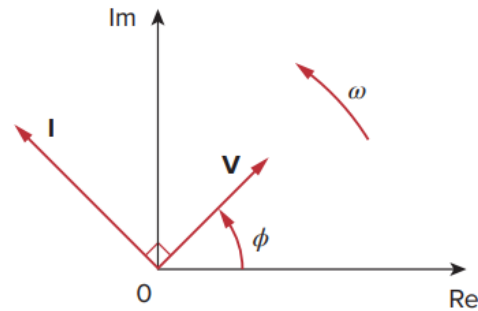


Figure 9.14

Phasor diagram for the capacitor; $\mathbf{I}$ leads $\mathbf{V}$.

3. For the capacitor C , assume the voltage across it is $v = V_m \cos(\omega t + \phi)$. The current through the capacitor is

$$i = C \frac{dv}{dt}$$

By following the same steps as we took for the inductor,

$$\mathbf{I} = j\omega C \mathbf{V} \quad \Rightarrow \quad \mathbf{V} = \frac{\mathbf{I}}{j\omega C}$$

4. The following table summarizes the phasor representation for circuit elements:

TABLE 9.2

Summary of voltage-current relationships.

Element	Time domain	Frequency domain
R	$v = Ri$	$\mathbf{V} = R\mathbf{I}$
L	$v = L \frac{di}{dt}$	$\mathbf{V} = j\omega L \mathbf{I}$
C	$i = C \frac{dv}{dt}$	$\mathbf{V} = \frac{\mathbf{I}}{j\omega C}$

1.1.4 Impedance and Admittance

1. The **impedance** $\mathbf{Z}$ of a circuit is the ratio of the phasor voltage $\mathbf{V}$ to the phasor current $\mathbf{I}$, measured in ohms (Ω). The impedance represents the opposition that the circuit exhibits to the flow of sinusoidal current.

$$\mathbf{Z} = \frac{\mathbf{V}}{\mathbf{I}}, \mathbf{V} = \mathbf{I}\mathbf{Z}$$

2. The **admittance** $\mathbf{Y}$ is the reciprocal of impedance, measured in siemens (S).

$$\mathbf{Y} = \frac{1}{\mathbf{Z}} = \frac{\mathbf{I}}{\mathbf{V}}, \mathbf{I} = \mathbf{Y}\mathbf{V}$$

3. Although the impedance and admittance are the ratio of two phasors, they are **not** phasors, because they don't correspond to a sinusoidally varying quantity.
4. The table 9.3 summarizes the impedance and admittance of three passive circuit elements. Fig. 9.15 shows the limiting cases when $\omega \rightarrow 0$ (DC limit) or $\omega \rightarrow \infty$ (high frequency limit).

TABLE 9.3		
Impedances and admittances of passive elements.		
Element	Impedance	Admittance
R	$Z = R$	$Y = \frac{1}{R}$
L	$Z = j\omega L$	$Y = \frac{1}{j\omega L}$
C	$Z = \frac{1}{j\omega C}$	$Y = j\omega C$

(a)

(b)

Figure 9.15
Equivalent circuits at dc and high frequencies: (a) inductor. (b) capacitor.

Figure 9.15

Equivalent circuits at dc and high frequencies: (a) inductor, (b) capacitor.

5. As a complex quantity, the impedance may be expressed in rectangular form as

$$\mathbf{Z} = R \pm jX \quad (9.41)$$

where $R = \text{Re } \mathbf{Z}$ is the *resistance* and $X = \text{Im } \mathbf{Z}$ is the *reactance*.

The reactance, X , is just a magnitude, a positive value, but when used as a vector, a j is associated with inductance and a $-j$ is associated with capacitance. Thus, impedance $\mathbf{Z} = R + jX$ is said to be *inductive* or lagging since current lags voltage, while impedance $\mathbf{Z} = R - jX$ is *capacitive* or leading because current leads voltage. The impedance, resistance, and reactance are all measured in ohms. The impedance may also be expressed in polar form as

$$\mathbf{Z} = |\mathbf{Z}| \angle \theta$$

6. As a complex quantity, we may write $\mathbf{Y}$ as

$$\mathbf{Y} = G + jB$$

where $G = \text{Re } \mathbf{Y}$ is called the *conductance* and $B = \text{Im } \mathbf{Y}$ is called the *susceptance*. Admittance, conductance, and susceptance are all expressed in the unit of siemens (or mhos).

1.1.5 Kirchhoff's law in the frequency domain

1. For KVL, let $v_1, v_2, \dots, v_n$ be voltages around a closed loop. Then

$$v_1 + v_2 + \dots + v_n = 0$$

In sinusoidal steady state,

$$0 = V_1 \cos(\omega t + \phi_1) + V_2 \cos(\omega t + \phi_2) + \dots + V_n \cos(\omega t + \phi_n)$$

$$0 = \text{Re} [V_1 e^{j(\omega t + \phi_1)} + V_2 e^{j(\omega t + \phi_2)} + \dots + V_n e^{j(\omega t + \phi_n)}]$$

$$0 = \text{Re}[(\mathbf{V}_1 + \mathbf{V}_2 + \dots + \mathbf{V}_n) e^{j\omega t}]$$

Because $e^{j\omega t} \neq 0$,

$$\mathbf{0} = \mathbf{V}_1 + \mathbf{V}_2 + \dots + \mathbf{V}_n$$

which states that Kirchhoff's voltage law holds for phasors.

2. Let $i_1, i_2, \dots, i_n$ be the current leaving or entering a closed surface in a network at time t , then KCL implies

$$i_1 + i_2 + \dots + i_n = 0$$

Applying the similar procedure as KVL for phasors, we can show that

$$\mathbf{I}_1 + \mathbf{I}_2 + \dots + \mathbf{I}_n = 0$$

which states that Kirchhoff's current law holds for phasors.

3. The fact that Kirchhoff's law holds in the frequency domain forms the basis of circuit simplification techniques (source transformation, impedance combination, wye-delta transformation) and theorem (Thevenin theorem, Norton theorem) extended for phasors.

1.1.6 Impedance simplification

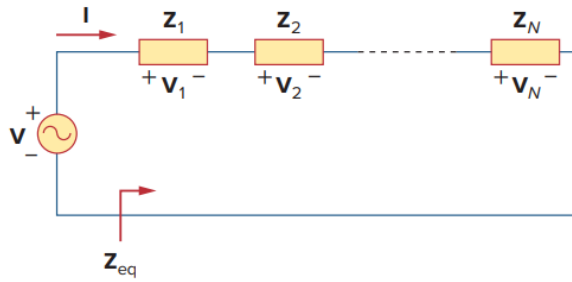


Figure 9.18
 N impedances in series.

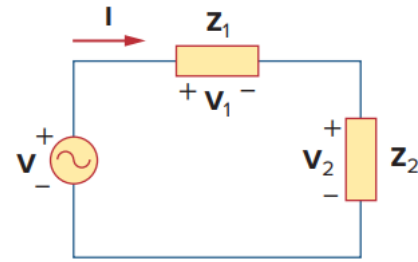


Figure 9.19
Voltage division.

1. Consider the N series-connected impedances shown in Fig. 9.18. The same current $\mathbf{I}$ flows through the impedances. Applying KVL around the loop gives

$$\mathbf{V} = \mathbf{I}(\mathbf{Z}_1 + \mathbf{Z}_2 + \cdots + \mathbf{Z}_n)$$

The equivalent impedance at the input terminals is

$$\mathbf{Z}_{\text{eq}} = \mathbf{Z}_1 + \mathbf{Z}_2 + \cdots + \mathbf{Z}_n$$

2. If $N = 2$, as shown in Fig. 9.19, the current through the impedances is

$$\mathbf{I} = \frac{\mathbf{V}}{\mathbf{Z}_1 + \mathbf{Z}_2}$$

Because $\mathbf{V}_1 = \mathbf{I}\mathbf{Z}_1$, $\mathbf{V}_2 = \mathbf{I}\mathbf{Z}_2$,

$$\mathbf{V}_1 = \frac{\mathbf{Z}_1}{\mathbf{Z}_1 + \mathbf{Z}_2} \mathbf{V}, \mathbf{V}_2 = \frac{\mathbf{Z}_2}{\mathbf{Z}_1 + \mathbf{Z}_2} \mathbf{V}$$

which is the voltage division relationship.

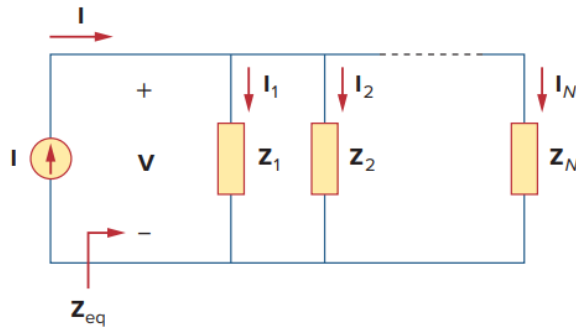


Figure 9.20
N impedances in parallel.

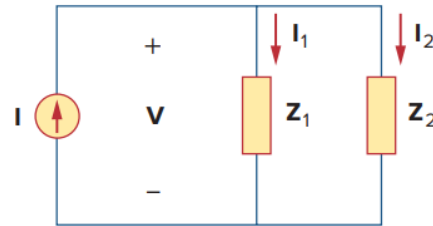


Figure 9.21
Current division.

3. In the same manner, we can obtain the equivalent impedance or admittance of the N parallel-connected impedances shown in Fig. 9.20. The voltage across each impedance is the same. Applying KCL at the top node,

$$I = I_1 + I_2 + \cdots + I_n = V \left(\frac{1}{Z_1} + \frac{1}{Z_2} + \cdots + \frac{1}{Z_n} \right)$$

The equivalent impedance at the input terminals is

$$\frac{1}{Z_{eq}} = \left(\frac{1}{Z_1} + \frac{1}{Z_2} + \cdots + \frac{1}{Z_n} \right)$$

and the equivalent admittance becomes

$$Y_{eq} = Y_1 + Y_2 + \cdots + Y_n$$

4. Similarly, the current division relationship is

$$I_1 = \frac{Y_1}{Y_1 + Y_2} I, I_2 = \frac{Y_2}{Y_1 + Y_2} I$$

5. The delta-to-wye and wye-to-delta transformations that we applied to resistive circuits are also valid for impedances. With reference to Fig. 9.22, the conversion formulas are as follows.

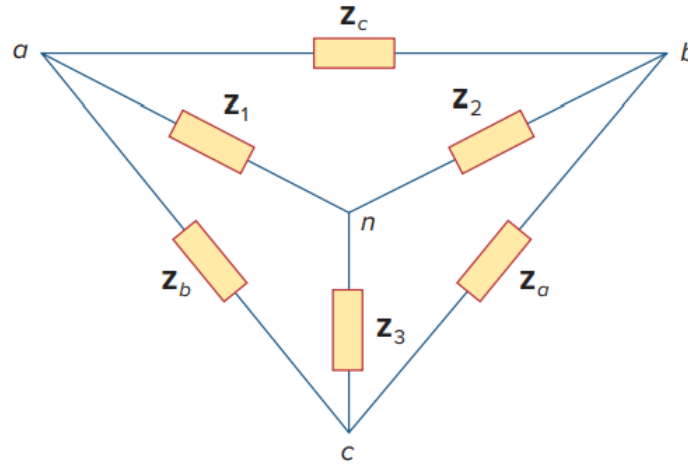


Figure 9.22
Superimposed Y and Δ networks.

Y - Δ Conversion: "Divide the sum of the products of all Y resistors by the value of the opposing Y resistor."

$$Z_a = \frac{Z_1 Z_2 + Z_2 Z_3 + Z_3 Z_1}{Z_1}$$

$$Z_b = \frac{Z_1 Z_2 + Z_2 Z_3 + Z_3 Z_1}{Z_2}$$

$$Z_c = \frac{Z_1 Z_2 + Z_2 Z_3 + Z_3 Z_1}{Z_3}$$

Δ - Y Conversion: "Multiply the two adjacent delta resistances on either side of the Y resistor you are determining, and divide by the sum of all delta resistances."

$$Z_1 = \frac{Z_b Z_c}{Z_a + Z_b + Z_c}$$

$$Z_2 = \frac{Z_c Z_a}{Z_a + Z_b + Z_c}$$

$$Z_3 = \frac{Z_a Z_b}{Z_a + Z_b + Z_c}$$

When a Δ - Y circuit is balanced, Eqs. (9.67) and (9.68) become

$$Z_{\Delta} = 3Z_Y \quad \text{or} \quad Z_Y = \frac{1}{3}Z_{\Delta} \quad (9.69)$$

where $Z_Y = Z_1 = Z_2 = Z_3$ and $Z_{\Delta} = Z_a = Z_b = Z_c$.

6. **Example.** Find current $\mathbf{I}$ in the circuit of Fig. 9.28.

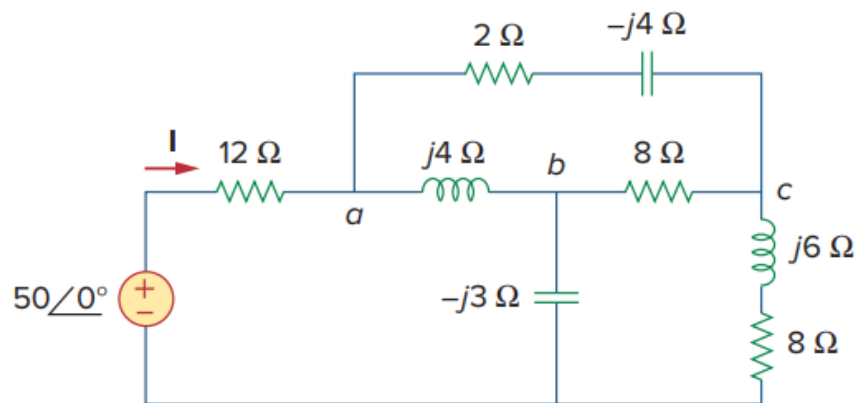
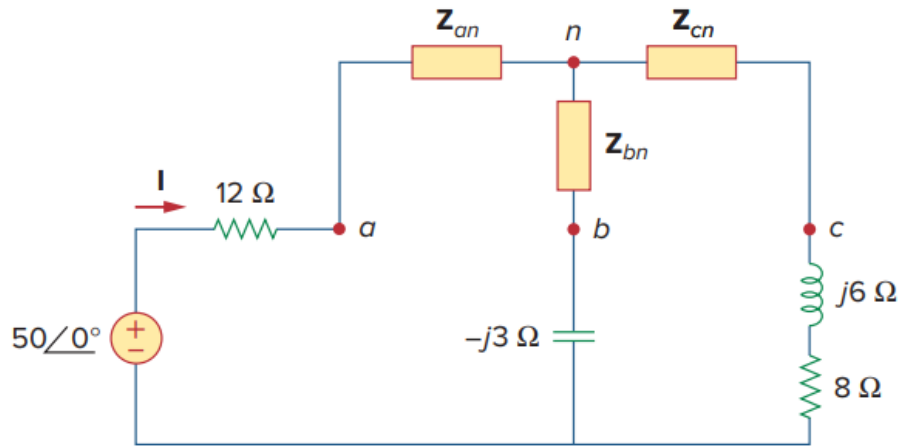


Figure 9.28
For Example 9.12.

**Figure 9.29**

The circuit in Fig. 9.28 after delta-to-wye transformation.

Solution:

The delta network connected to nodes a , b , and c can be converted to the Y network of Fig. 9.29. We obtain the Y impedances as follows using Eq. (9.68):

$$Z_{an} = \frac{j4(2 - j4)}{j4 + 2 - j4 + 8} = \frac{4(4 + j2)}{10} = (1.6 + j0.8) \Omega$$

$$Z_{bn} = \frac{j4(8)}{10} = j3.2 \Omega, \quad Z_{cn} = \frac{8(2 - j4)}{10} = (1.6 - j3.2) \Omega$$

The total impedance at the source terminals is

$$\begin{aligned} \mathbf{Z} &= 12 + Z_{an} + (Z_{bn} - j3) \parallel (Z_{cn} + j6 + 8) \\ &= 12 + 1.6 + j0.8 + (j0.2) \parallel (9.6 + j2.8) \\ &= 13.6 + j0.8 + \frac{j0.2(9.6 + j2.8)}{9.6 + j3} \\ &= 13.6 + j1 = 13.64 \angle 4.204^\circ \Omega \end{aligned}$$

The desired current is

$$\mathbf{I} = \frac{\mathbf{V}}{\mathbf{Z}} = \frac{50 \angle 0^\circ}{13.64 \angle 4.204^\circ} = 3.666 \angle -4.204^\circ \text{ A}$$

1.2 Sinusoidal Steady-State analysis

Steps to Analyze AC Circuits:

1. Transform the circuit to the phasor or frequency domain.
2. Solve the problem using circuit techniques (nodal analysis, mesh analysis, superposition, etc.).
3. Transform the resulting phasor to the time domain.

1.2.1 Nodal analysis in frequency domain

- **Example 1. Nodal analysis in frequency domain.** Find i_x in the circuit of Fig. 10.1 using nodal analysis.

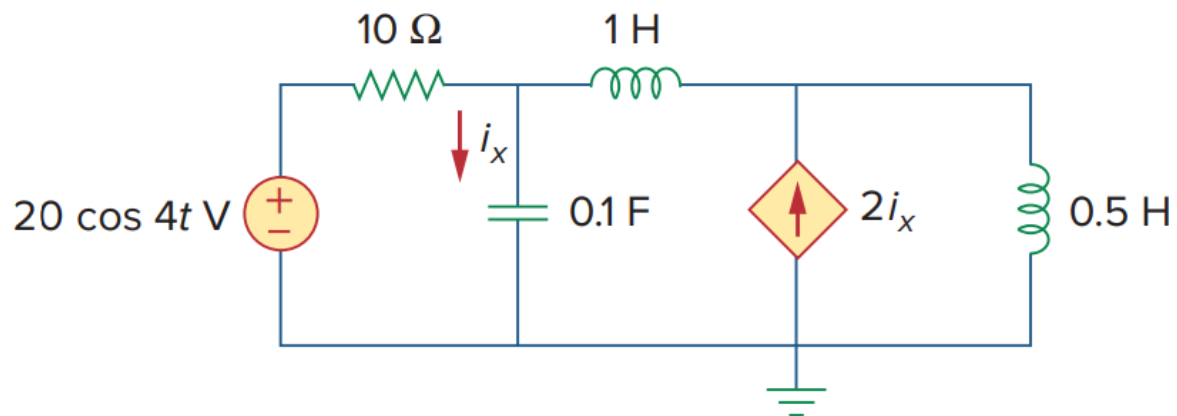


Figure 10.1
For Example 10.1.

Solution:

We first convert the circuit to the frequency domain:

$$\begin{aligned} 20 \cos 4t &\Rightarrow 20\angle 0^\circ, \quad \omega = 4 \text{ rad/s} \\ 1 \text{ H} &\Rightarrow j\omega L = j4 \\ 0.5 \text{ H} &\Rightarrow j\omega L = j2 \\ 0.1 \text{ F} &\Rightarrow \frac{1}{j\omega C} = -j2.5 \end{aligned}$$

Thus, the frequency domain equivalent circuit is as shown in Fig. 10.2.

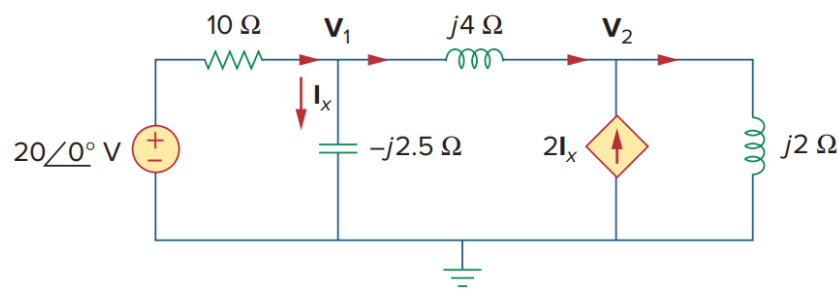


Figure 10.2

Frequency domain equivalent of the circuit in Fig. 10.1.

Applying KCL at node 1,

$$\frac{20 - V_1}{10} = \frac{V_1}{-j2.5} + \frac{V_1 - V_2}{j4}$$

or

$$(1 + j1.5)V_1 + j2.5V_2 = 20 \quad (10.1.1)$$

At node 2,

$$2I_x + \frac{V_1 - V_2}{j4} = \frac{V_2}{j2}$$

But $I_x = \frac{V_1}{-j2.5}$. Substituting this gives

$$\frac{2V_1}{-j2.5} + \frac{V_1 - V_2}{j4} = \frac{V_2}{j2}$$

By simplifying, we get

$$11V_1 + 15V_2 = 0 \quad (10.1.2)$$

Equations (10.1.1) and (10.1.2) can be put in matrix form as

$$\begin{bmatrix} 1 + j1.5 & j2.5 \\ 11 & 15 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} 20 \\ 0 \end{bmatrix}$$

We obtain the determinants as

$$\Delta = \begin{vmatrix} 1 + j1.5 & j2.5 \\ 11 & 15 \end{vmatrix} = 15 - j5$$

$$\Delta_1 = \begin{vmatrix} 20 & j2.5 \\ 0 & 15 \end{vmatrix} = 300, \quad \Delta_2 = \begin{vmatrix} 1 + j1.5 & 20 \\ 11 & 0 \end{vmatrix} = -220$$

$$V_1 = \frac{\Delta_1}{\Delta} = \frac{300}{15 - j5} = 18.97 \angle 18.43^\circ \text{ V}$$

$$V_2 = \frac{\Delta_2}{\Delta} = \frac{-220}{15 - j5} = 13.91 \angle 198.3^\circ \text{ V}$$

The current I_x is given by

$$I_x = \frac{V_1}{-j2.5} = \frac{18.97 \angle 18.43^\circ}{2.5 \angle -90^\circ} = 7.59 \angle 108.4^\circ \text{ A}$$

Transforming this to the time domain,

$$i_x = 7.59 \cos(4t + 108.4^\circ) \text{ A}$$

- **Example 2. Supernode treatment** Compute V_1 and V_2 in Fig. 10.4

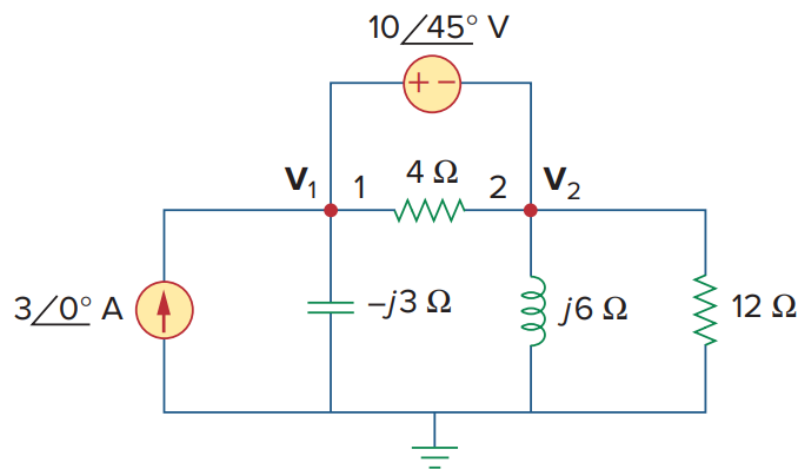
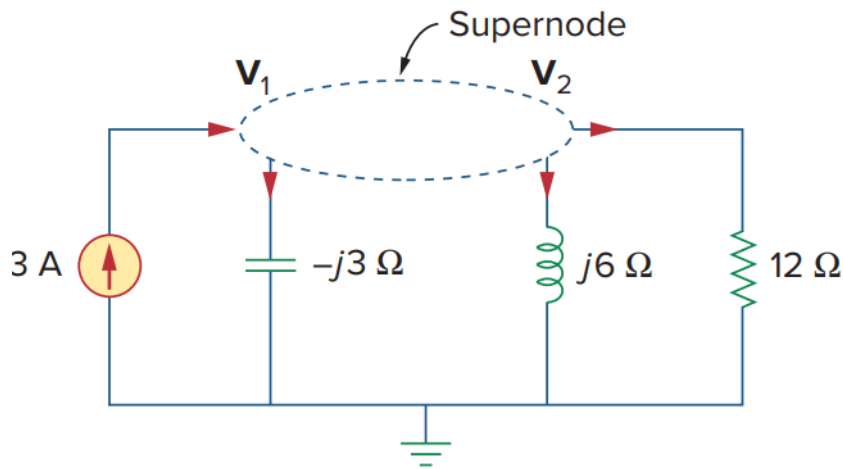


Figure 10.4
For Example 10.2.

Solution:

Nodes 1 and 2 form a supernode as shown in Fig. 10.5. Applying KCL at the

**Figure 10.5**

A supernode in the circuit of Fig. 10.4.

supernode gives

$$3 = \frac{V_1}{-j3} + \frac{V_2}{j6} + \frac{V_2}{12}$$

or

$$36 = j4V_1 + (1 - j2)V_2 \quad (10.2.1)$$

But a voltage source is connected between nodes 1 and 2, so that

$$V_1 = V_2 + 10\angle 45^\circ \quad (10.2.2)$$

Substituting Eq. (10.2.2) in Eq. (10.2.1) results in

$$36 - 40\angle 135^\circ = (1 + j2)V_2 \Rightarrow V_2 = 31.41\angle -87.18^\circ \text{ V}$$

From Eq. (10.2.2),

$$V_1 = V_2 + 10\angle 45^\circ = 25.78\angle -70.48^\circ \text{ V}$$

1.2.2 Mesh analysis in frequency domain

- **Example 1: Mesh analysis in the frequency domain**

Determine current I_o in the circuit of Fig. 10.7 using mesh analysis.

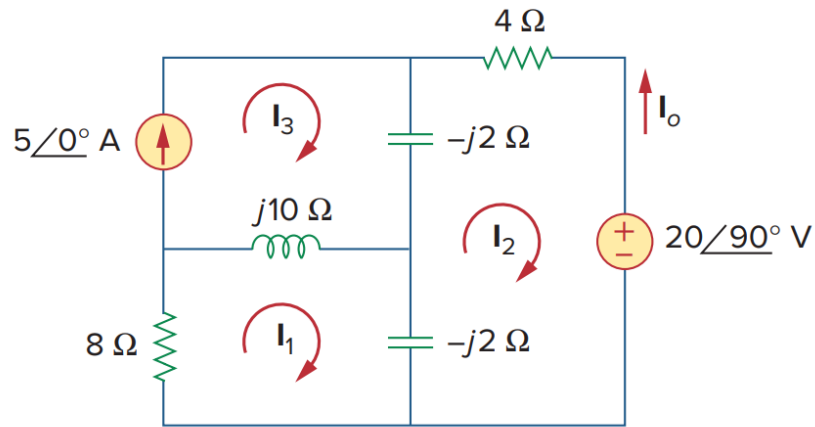


Figure 10.7

For Example 10.3.

Solution:

Applying KVL to mesh 1, we obtain

$$(8 + j10 - j2)I_1 - (-j2)I_2 - j10I_3 = 0 \quad (10.3.1)$$

For mesh 2,

$$(4 - j2 - j2)I_2 - (-j2)I_1 - (-j2)I_3 + 20\angle 90^\circ = 0 \quad (10.3.2)$$

For mesh 3, $I_3 = 5$. Substituting this in Eqs. (10.3.1) and (10.3.2), we get

$$(8 + j8)I_1 + j2I_2 = j50 \quad (10.3.3)$$

$$j2I_1 + (4 - j4)I_2 = -j20 - j10 \quad (10.3.4)$$

Equations (10.3.3) and (10.3.4) can be put in matrix form as

$$\begin{bmatrix} 8 + j8 & j2 \\ j2 & 4 - j4 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} j50 \\ -j30 \end{bmatrix}$$

from which we obtain the determinants

$$\Delta = \begin{vmatrix} 8 + j8 & j2 \\ j2 & 4 - j4 \end{vmatrix} = 32(1 + j)(1 - j) + 4 = 68$$

$$\Delta_2 = \begin{vmatrix} 8 + j8 & j50 \\ j2 & -j30 \end{vmatrix} = 340 - j240 = 416.17\angle -35.22^\circ$$

$$I_2 = \frac{\Delta_2}{\Delta} = \frac{416.17\angle -35.22^\circ}{68} = 6.12\angle -35.22^\circ \text{ A}$$

The desired current is

$$I_o = -I_2 = 6.12\angle 144.78^\circ \text{ A}$$

- **Example 2: Supermesh treatment in the frequency domain**

Solve for V_o in the circuit of Fig. 10.9 using mesh analysis.

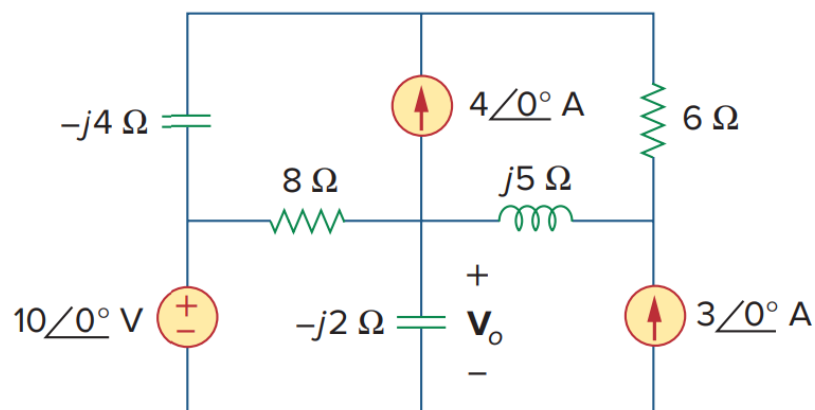
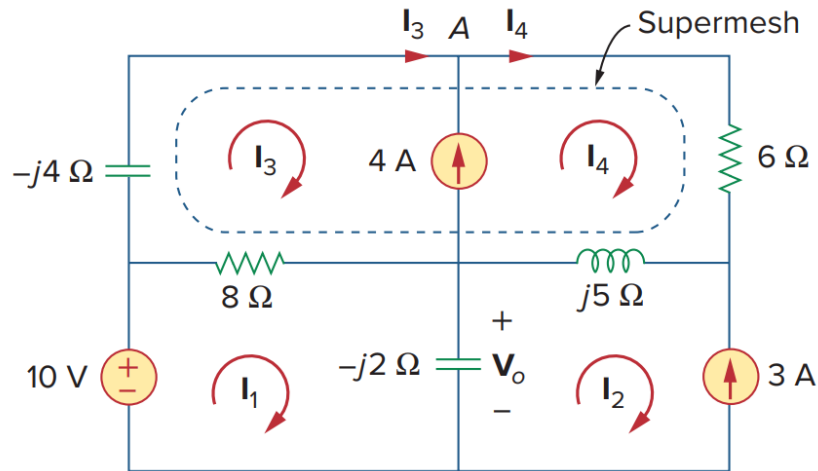


Figure 10.9
For Example 10.4.

**Figure 10.10**

Analysis of the circuit in Fig. 10.9.

Solution:

As shown in Fig. 10.10, meshes 3 and 4 form a supermesh due to the current source between the meshes. For mesh 1, KVL gives

$$-10 + (8 - j2)I_1 - (-j2)I_2 - 8I_3 = 0$$

or

$$(8 - j2)I_1 + j2I_2 - 8I_3 = 10 \quad (10.4.1)$$

For mesh 2,

$$I_2 = -3 \quad (10.4.2)$$

For the supermesh,

$$(8 - j4)I_3 - 8I_1 + (6 + j5)I_4 - j5I_2 = 0 \quad (10.4.3)$$

Due to the current source between meshes 3 and 4, at node A,

$$I_4 = I_3 + 4 \quad (10.4.4)$$

Instead of solving the above four equations, we reduce them to two by elimination.

Combining Eqs. (10.4.1) and (10.4.2),

$$(8 - j2)I_1 - 8I_3 = 10 + j6 \quad (10.4.5)$$

Combining Eqs. (10.4.2) to (10.4.4),

$$-8I_1 + (14 + j)I_3 = -24 - j35 \quad (10.4.6)$$

From Eqs. (10.4.5) and (10.4.6), we obtain the matrix equation

$$\begin{bmatrix} 8 - j2 & -8 \\ -8 & 14 + j \end{bmatrix} \begin{bmatrix} I_1 \\ I_3 \end{bmatrix} = \begin{bmatrix} 10 + j6 \\ -24 - j35 \end{bmatrix}$$

We obtain the following determinants

$$\Delta = \begin{vmatrix} 8 - j2 & -8 \\ -8 & 14 + j \end{vmatrix} = 112 + j8 - j28 + 2 - 64 = 50 - j20$$

$$\Delta_1 = \begin{vmatrix} 10 + j6 & -8 \\ -24 - j35 & 14 + j \end{vmatrix} = 140 + j10 + j84 - 6 - 192 - j280 = -58 - j186$$

Current I_1 is obtained as

$$I_1 = \frac{\Delta_1}{\Delta} = \frac{-58 - j186}{50 - j20} = 3.618 \angle 274.5^\circ \text{ A}$$

The required voltage V_o is

$$\begin{aligned} V_o &= -j2(I_1 - I_2) = -j2(3.618 \angle 274.5^\circ + 3) \\ &= -7.2134 - j6.568 = 9.756 \angle 222.32^\circ \text{ V} \end{aligned}$$

1.2.3 Superposition theorem

Since ac circuits are **linear**, the superposition theorem applies to ac circuits the same way it applies to dc circuits. The theorem becomes important if the circuit has sources operating at **different frequencies**. In this case,

1. Since the impedances depend on frequency, we must have a different frequency domain circuit for each frequency. The total response must be obtained by adding the individual responses in the time domain.
2. It is incorrect to try to add the responses in the phasor or frequency domain. Why? Because the exponential factor $e^{j\omega t}$ is implicit in sinusoidal analysis, and that factor would change for every angular frequency ω . It would therefore not make sense to add responses at different frequencies in the phasor domain.

Thus, when a circuit has sources operating at different frequencies, one must add the responses due to the individual frequencies in the **time domain**.

Example: AC sources operating at different frequencies.

Find v_o of the circuit of Fig. 10.13 using the superposition theorem.

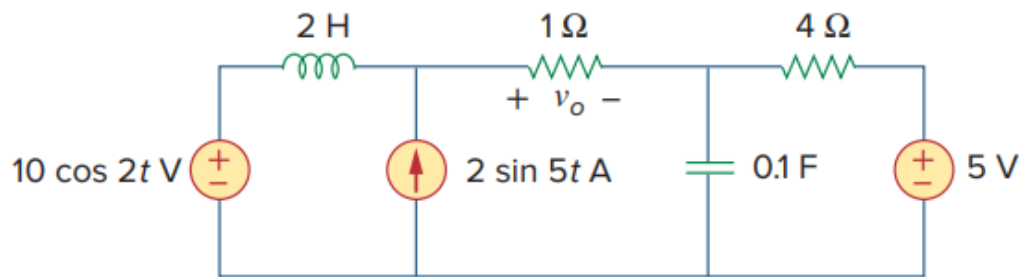
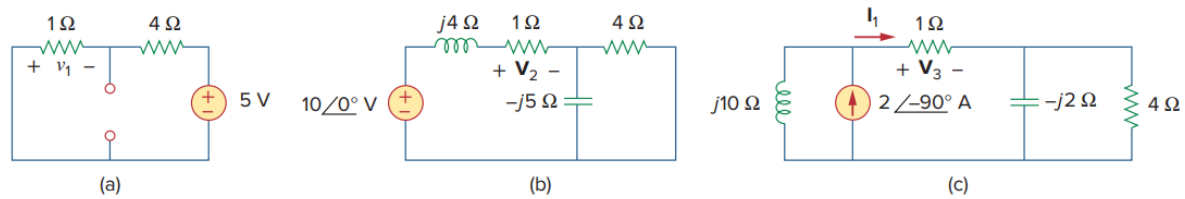


Figure 10.13
For Example 10.6.

**Figure 10.14**

Solution of Example 10.6: (a) setting all sources to zero except the 5-V dc source, (b) setting all sources to zero except the ac voltage source, (c) setting all sources to zero except the ac current source.

Solution:

Because the circuit operates at three different frequencies ($\omega = 0$ for the dc voltage source), one way to obtain a solution is to use superposition, which breaks the problem into single-frequency problems. So we let

$$v_o = v_1 + v_2 + v_3 \quad (10.6.1)$$

where v_1 is due to the 5-V dc voltage source, v_2 is due to the $10 \cos 2t$ V voltage source, and v_3 is due to the $2 \sin 5t$ A current source.

To find v_1 , we set to zero all sources except the 5-V dc source. We recall that at steady state, a capacitor is an open circuit to dc while an inductor is a short circuit to dc. There is an alternative way of looking at this. Because $\omega = 0$, $j\omega L = 0$, $1/j\omega C = \infty$. Either way, the equivalent circuit is as shown in Fig. 10.14(a). By voltage division,

$$-v_1 = \frac{1}{1+4}(5) = 1 \text{ V} \quad (10.6.2)$$

To find v_2 , we set to zero both the 5-V source and the $2 \sin 5t$ current source and transform the circuit to the frequency domain.

$$10 \cos 2t \Rightarrow 10\angle 0^\circ, \quad \omega = 2 \text{ rad/s}$$

$$2 \text{ H} \Rightarrow j\omega L = j4 \Omega$$

$$0.1 \text{ F} \Rightarrow \frac{1}{j\omega C} = -j5 \Omega$$

The equivalent circuit is now as shown in Fig. 10.14(b). Let

$$Z = -j5 \parallel 4 = \frac{-j5 \times 4}{4 - j5} = 2.439 - j1.951$$

By voltage division,

$$V_2 = \frac{1}{1 + j4 + Z}(10\angle 0^\circ) = \frac{10}{3.439 + j2.049} = 2.498\angle -30.79^\circ$$

In the time domain,

$$v_2 = 2.498 \cos(2t - 30.79^\circ) \quad (10.6.3)$$

To obtain v_3 , we set the voltage sources to zero and transform what is left to the frequency domain.

$$2 \sin 5t \Rightarrow 2\angle -90^\circ, \quad \omega = 5 \text{ rad/s}$$

$$2 \text{ H} \Rightarrow j\omega L = j10 \Omega$$

$$0.1 \text{ F} \Rightarrow \frac{1}{j\omega C} = -j2 \Omega$$

The equivalent circuit is in Fig. 10.14(c). Let

$$Z_1 = -j2 \parallel 4 = \frac{-j2 \times 4}{4 - j2} = 0.8 - j1.6 \, \Omega$$

By current division,

$$I_1 = \frac{j10}{j10 + 1 + Z_1} (2\angle -90^\circ) \text{ A}$$

$$V_3 = I_1 \times 1 = \frac{j10}{1.8 + j8.4} (-j2) = 2.328\angle -80^\circ \text{ V}$$

In the time domain,

$$v_3 = 2.33 \cos(5t - 80^\circ) = 2.33 \sin(5t + 10^\circ) \text{ V} \quad (10.6.4)$$

Substituting Eqs. (10.6.2) to (10.6.4) into Eq. (10.6.1), we have

$$v_o(t) = -1 + 2.498 \cos(2t - 30.79^\circ) + 2.33 \sin(5t + 10^\circ) \text{ V}$$

1.2.4 Source Transformation

As Fig. 10.16 shows, source transformation in the frequency domain involves transforming a voltage source in series with an impedance to a current source in parallel with an impedance, or vice versa.

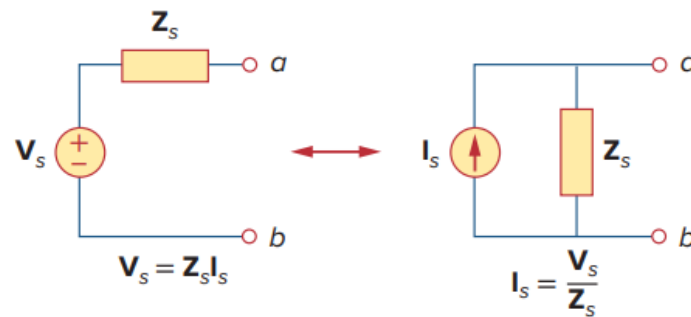


Figure 10.16
Source transformation.

As we go from one source type to another, we must keep the following relationship in mind:

$$\boxed{\mathbf{V}_s = \mathbf{Z}_s \mathbf{I}_s \quad \Longleftrightarrow \quad \mathbf{I}_s = \frac{\mathbf{V}_s}{\mathbf{Z}_s}}$$

Example: Calculate $\mathbf{V}_x$ in the circuit of Fig. 10.17 using the method of source transformation.

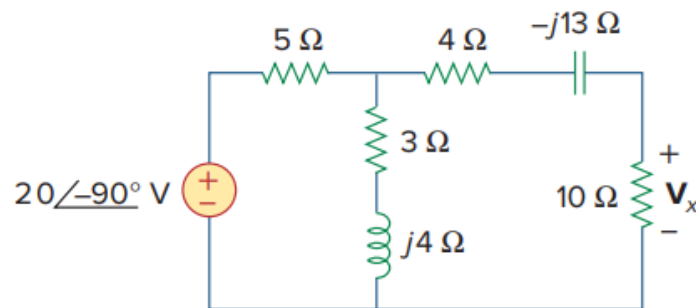
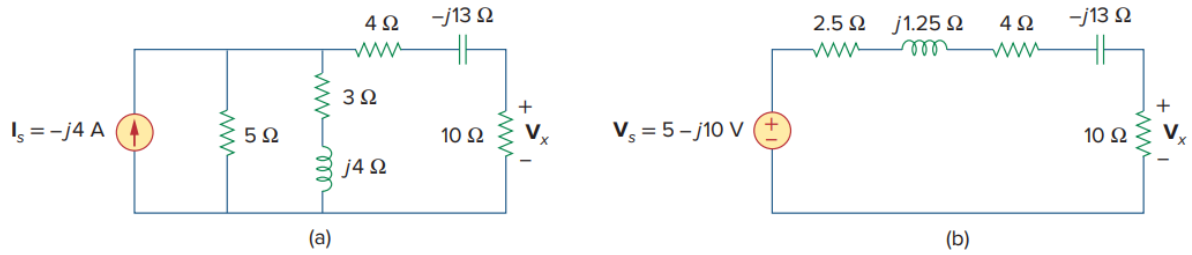


Figure 10.17
For Example 10.7.

**Figure 10.18**

Solution of the circuit in Fig. 10.17.

Solution:

We transform the voltage source to a current source and obtain the circuit in Fig. 10.18(a), where

$$\mathbf{I}_s = \frac{20/\angle -90^\circ}{5} = \frac{4}{\angle -90^\circ} = -j4 \text{ A}$$

The parallel combination of 5- Ω resistance and $(3 + j4)$ impedance gives

$$\mathbf{Z}_1 = \frac{5(3 + j4)}{8 + j4} = 2.5 + j1.25 \text{ } \Omega$$

Converting the current source to a voltage source yields the circuit in Fig. 10.18(b), where

$$\mathbf{V}_s = \mathbf{I}_s \mathbf{Z}_1 = (-j4)(2.5 + j1.25) = 5 - j10 \text{ V}$$

By voltage division,

$$\mathbf{V}_x = \frac{10}{10 + 2.5 + j1.25 + 4 - j13} (5 - j10) = 5.519/\angle -28^\circ \text{ V}$$

1.2.5 Thevenin and Norton equivalent

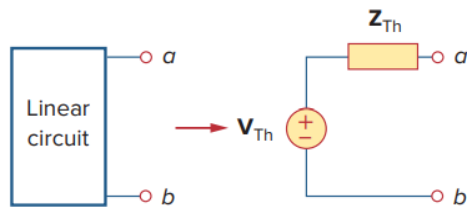


Figure 10.20
Thevenin equivalent.

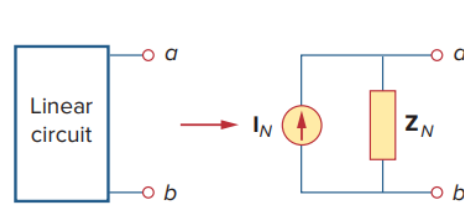


Figure 10.21
Norton equivalent.

Thevenin's and Norton's theorems are applied to ac circuits in the same way as they are to dc circuits. The only additional effort arises from the need to manipulate complex numbers. The frequency domain version of a Thevenin equivalent circuit is depicted in Fig. 10.20, where a linear circuit is replaced by a voltage source in series with an impedance. The Norton equivalent circuit is illustrated in Fig. 10.21, where a linear circuit is replaced by a current source in parallel with an impedance. Keep in mind that the two equivalent circuits are related as

$$\mathbf{V}_{Th} = \mathbf{Z}_N \mathbf{I}_N, \quad \mathbf{Z}_{Th} = \mathbf{Z}_N$$

just as in source transformation. $\mathbf{V}_{Th}$ is the open-circuit voltage while $\mathbf{I}_N$ is the short-circuit current.

Remark. If the circuit has sources operating at different frequencies, the Thevenin or Norton equivalent circuit must be determined at **each** frequency. This leads to entirely different equivalent circuits, one for each frequency, not one equivalent circuit with equivalent sources and equivalent impedances.

Example : Find the Thevenin equivalent of the circuit in Fig. 10.25 as seen from terminals a-b.

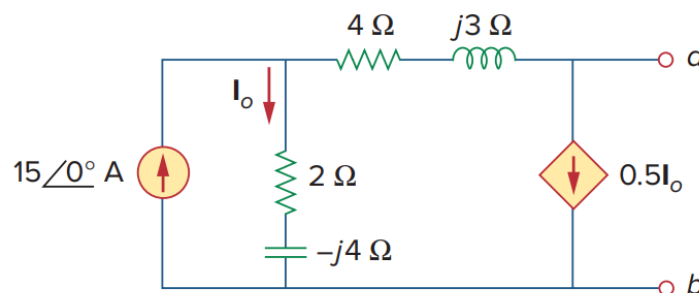
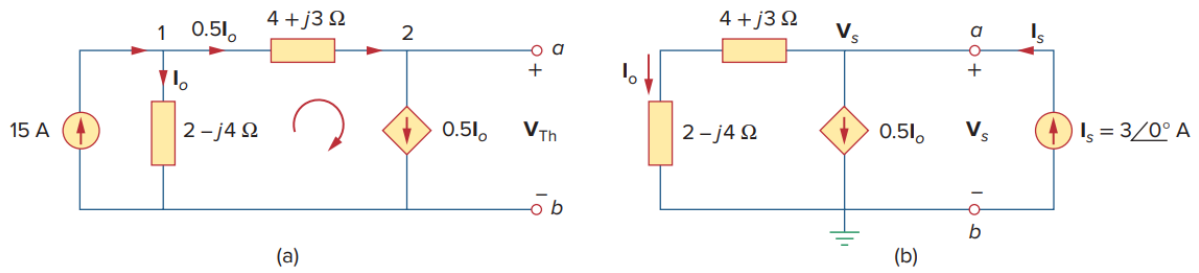


Figure 10.25
For Example 10.9.

**Figure 10.26**

Solution of the problem in Fig. 10.25: (a) finding V_{Th} , (b) finding Z_{Th} .

Solution:

To find V_{Th} , we apply KCL at node 1 in Fig. 10.26(a).

$$15 = I_0 + 0.5I_0 \Rightarrow I_0 = 10 \text{ A}$$

Applying KVL to the loop on the right-hand side in Fig. 10.26(a), we obtain

$$-I_0(2 - j4) + 0.5I_0(4 + j3) + V_{Th} = 0$$

or

$$V_{Th} = 10(2 - j4) - 5(4 + j3) = -j55$$

Thus, the Thevenin voltage is

$$V_{Th} = 55/\angle -90^\circ \text{ V}$$

To obtain Z_{Th} , we remove the independent source. Due to the presence of the dependent current source, we connect a 3-A current source (3 is an arbitrary value chosen for convenience here, a number divisible by the sum of currents leaving the node) to terminals a - b as shown in Fig. 10.26(b). At the node, KCL gives

$$3 = I_0 + 0.5I_0 \Rightarrow I_0 = 2 \text{ A}$$

Applying KVL to the outer loop in Fig. 10.26(b) gives

$$V_s = I_0(4 + j3 + 2 - j4) = 2(6 - j)$$

The Thevenin impedance is

$$Z_{Th} = \frac{V_s}{I_s} = \frac{2(6 - j)}{3} = 4 - j0.6667 \Omega$$

Example: Obtain current $\mathbf{I_o}$ in Fig. 10.28 using Norton's theorem.

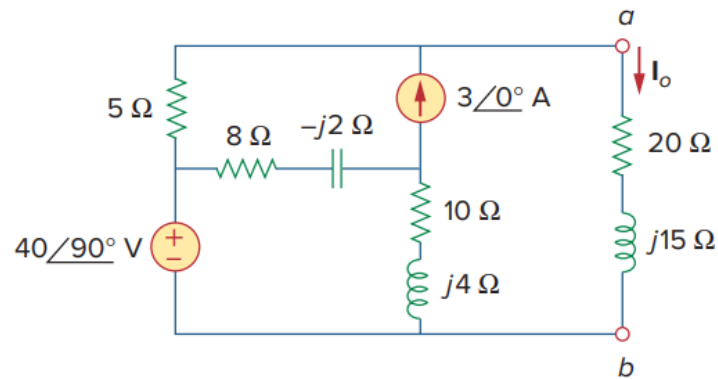
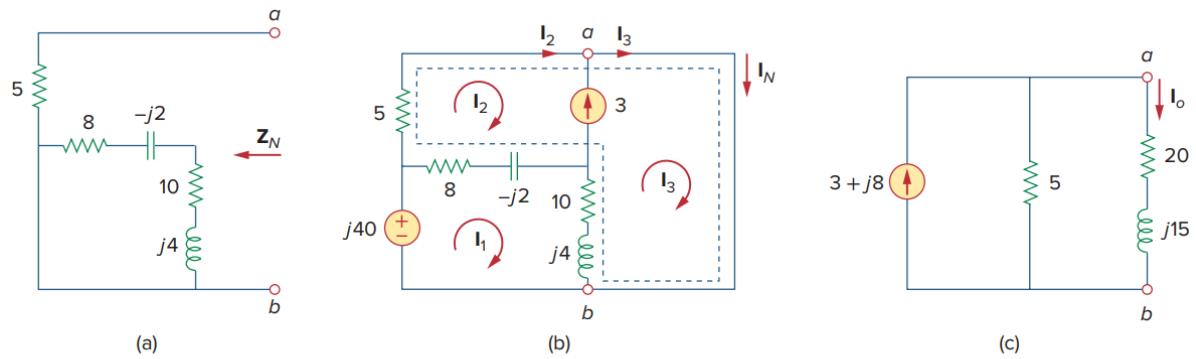


Figure 10.28
For Example 10.10.

**Figure 10.29**

Solution of the circuit in Fig. 10.28: (a) finding $\mathbf{Z}_N$, (b) finding $\mathbf{V}_N$, (c) calculating $\mathbf{I}_o$.

Solution:

Our first objective is to find the Norton equivalent at terminals a - b . $\mathbf{Z}_N$ is found in the same way as $\mathbf{Z}_{Th}$. We set the sources to zero as shown in Fig. 10.29(a). As evident from the figure, the $(8 - j2)$ and $(10 + j4)$ impedances are short-circuited, so that

$$\mathbf{Z}_N = 5 \, \Omega$$

To get $\mathbf{I}_N$, we short-circuit terminals a - b as in Fig. 10.29(b) and apply mesh analysis. Notice that meshes 2 and 3 form a supermesh because of the current source linking them. For mesh 1,

$$-j40 + (18 + j2)I_1 - (8 - j2)I_2 - (10 + j4)I_3 = 0 \quad (10.10.1)$$

For the supermesh,

$$(13 - j2)I_2 + (10 + j4)I_3 - (18 + j2)I_1 = 0 \quad (10.10.2)$$

At node a , due to the current source between meshes 2 and 3,

$$I_3 = I_2 + 3 \quad (10.10.3)$$

Adding Eqs. (10.10.1) and (10.10.2) gives

$$-j40 + 5I_2 = 0 \quad \Rightarrow \quad I_2 = j8$$

From Eq. (10.10.3),

$$I_3 = I_2 + 3 = 3 + j8$$

The Norton current is

$$\mathbf{I}_N = I_3 = (3 + j8) \text{ A}$$

Figure 10.29(c) shows the Norton equivalent circuit along with the impedance at terminals a - b . By current division,

$$I_o = \frac{5}{5 + 20 + j15} \mathbf{I}_N = \frac{3 + j8}{5 + j3} = 1.465 / \angle 38.48^\circ \text{ A}$$

1.2.6 Op amp AC circuit

Example: Determine $v_o(t)$ for the op amp circuit in Fig. 10.31(a) if $v_s = 3 \cos 1000t$ (V).

Solution:

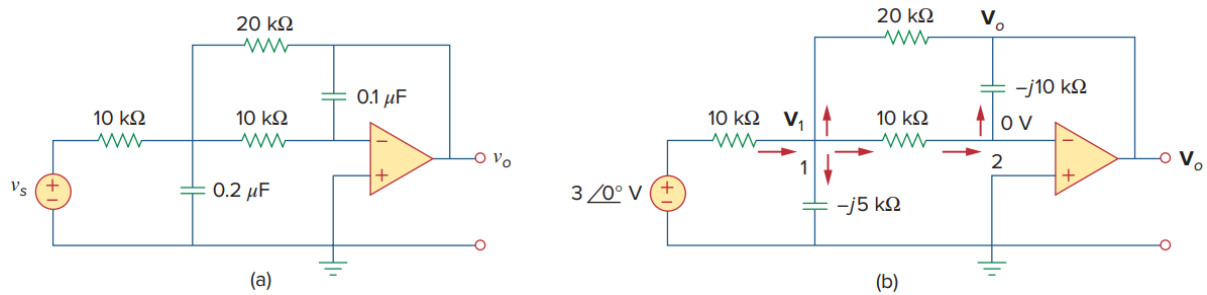


Figure 10.31

For Example 10.11: (a) the original circuit in the time domain, (b) its frequency domain equivalent.

We first transform the circuit to the frequency domain, as shown in Fig. 10.31(b), where $V_s = 3 \angle 0^\circ$, $\omega = 1000 \text{ rad/s}$. Applying KCL at node 1, we obtain

$$\frac{3 \angle 0^\circ - V_1}{10} = \frac{V_1}{-j5} + \frac{V_1 - 0}{10} + \frac{V_1 - V_o}{20}$$

or

$$6 = (5 + j4)V_1 - V_o \quad (10.11.1)$$

At node 2, KCL gives

$$\frac{V_1 - 0}{10} = \frac{0 - V_o}{-j10}$$

which leads to

$$V_1 = -jV_o \quad (10.11.2)$$

Substituting Eq. (10.11.2) into Eq. (10.11.1) yields

$$6 = -j(5 + j4)V_o - V_o = (3 - j5)V_o$$

$$V_o = \frac{6}{3 - j5} = 1.029 \angle 59.04^\circ$$

Hence,

$$v_o(t) = 1.029 \cos(1000t + 59.04^\circ) \text{ V}$$

Example: Compute the closed-loop gain and phase shift for the circuit in Fig. 10.33. Assume that $R_1 = R_2 = 10 \text{ k}\Omega$, $C_1 = 2 \text{ }\mu\text{F}$, $C_2 = 1 \text{ }\mu\text{F}$, and $\omega = 200 \text{ rad/s}$.

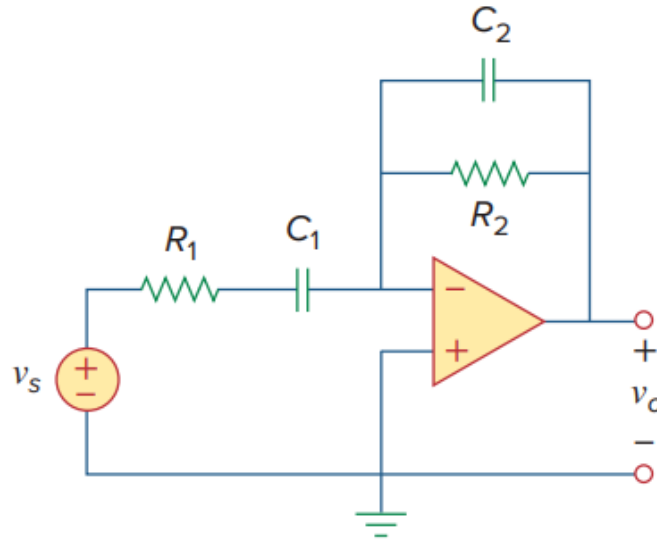


Figure 10.33
For Example 10.12.

Solution:

The feedback and input impedances are calculated as

$$\mathbf{Z}_f = R_2 \parallel \frac{1}{j\omega C_2} = \frac{R_2}{1 + j\omega R_2 C_2}$$

$$\mathbf{Z}_i = R_1 + \frac{1}{j\omega C_1} = \frac{1 + j\omega R_1 C_1}{j\omega C_1}$$

Since the circuit in Fig. 10.33 is an inverting amplifier, the closed-loop gain is given by

$$\mathbf{G} = \frac{V_o}{V_s} = -\frac{\mathbf{Z}_f}{\mathbf{Z}_i} = \frac{-j\omega C_1 R_2}{(1 + j\omega R_1 C_1)(1 + j\omega R_2 C_2)}$$

Substituting the given values of R_1 , R_2 , C_1 , C_2 , and ω , we obtain

$$\mathbf{G} = \frac{-j4}{(1 + j4)(1 + j2)} = 0.434/\angle 130.6^\circ$$

Thus, the closed-loop gain is 0.434 and the phase shift is 130.6° .

1.3 AC Power analysis (미작성)

1.4 Practical applications of AC circuits (미작성)